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Analysis of main constituents and mechanisms underlying antidepressant-like effects of Xiaochaihutang in mice



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CID: 547)

5-Hydroxytryptamine (PubChem CID: 5202)

4-Hydroxy-3-methoxyphenylacetic acid

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ABSTRACT

Ethnopharmacological relevance: Xiaochaihutang (XCHT), a famous Chinese herbal formula which consists of seven Chinese herbs, has been used clinically in depressive disorders in China. Our previous studies have demonstrated that XCHT improved depressive-like behavior in several animal models of depression. However, therapeutic basis of XCHT on depression are challenging, due to the complex active constituents of XCHT and the unclear pharmacological mechanism of action.

Materials and methods: To provide further insights into therapeutic basis of XCHT, the core in compatibility of XCHT on antidepressant therapy was assessed by the method of orthogonal array design. The comparative evaluations on antidepressant effects of XCHT and its core in compatibility were executed by tail suspension test (TST), forced swim test (FST), novelty suppressed feeding test (NSFT), reserpine-induced hypothermia and palpebral ptosis. Moreover, the potential mechanism was explored by investigating levels of monoamine neurotransmitters in hypothalamus and striatum and neurogenesis in hippocampus. Chemical profile of active constituents in plasma after oral administration of the core in compatibility of XCHT was revealed by ultra performance liquid chromatography-tandem mass spectrometry (UPLC–MS/MS).

Results: The results of orthogonal array design experiment showed that Huangqin (*Radix scutellariae*), Renshen (*Ginseng*) and Gancao (*Radix glycyrrhizae*), defined as HRG, might be the core in compatibility of XCHT on antidepressant therapy. In accordance with XCHT, oral administration of HRG for 15 days significantly reduced immobility duration in TST and FST without affecting locomotor activity. Both HRG and XCHT increased immobility latency in FST, decreased the latency in NSFT, reversed reserpine-induced hypothermia and palpebral ptosis. Moreover, both HRG and XCHT significantly increased levels of 5-HT and DA in hypothalamus. In addition, HRG could remarkably increase Ki-67 and doublecortin (DCX) positive cells in hippocampus. A total 25 active constituents in plasma, including 14 prototype components and 11 metabolites, were identified by UPLC–MS/MS after oral administration of HRG.

Conclusion: The present results reveal that HRG is supposed to be the core in compatibility of XCHT on antidepressant therapy. In accordance with XCHT, HRG exerts significant antidepressant-like effects, which are likely attributed to regulating serotonergic and dopaminergic systems and increasing hippocampal neurogenesis. The constituents identified in plasma after oral administration of HRG may be the potential active ingredients for the treatment of depression.

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Abbreviations: XCHT, Xiaochaihutang; TST, tail suspension test; FST, forced swimming test; NSF, novelty suppressed feeding; LAT, Locomotor activity test; TCM, traditional Chinese medicine; HRG, Huangqin (*Radix scutellariae*); Radix glycyrrhizae, Renshen (*Ginseng*) and Gancao; AMI, Amitriptyline; 5-HT, 5-Hydroxytryptamine; 5-HIAA, 5-Hydroxyindole-3-acetic acid; DA, Dopamine; (DOPAC), 3,4-Dihydroxyphenylacetic acid; HVA, 4-Hydroxy-3-methoxyphenylacetic acid

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1. Introduction

Depression is a common recurrent psychiatric disorder, which has high lifetime prevalence and severely affects quality of human life. Although classical antidepressants, such as selective serotonin reuptake inhibitors (SSRIs), have been widely used in clinic, these antidepressants are often accompanied with various side effects and were found effective in only 50% of the patients (Duman and

Aghajanian, 2012; Jia et al., 2009; Li and Gao, 1996). In China, traditional Chinese medicines (TCMs) exert excellent antidepressant effects with fewer side effects when used therapeutically (Freeman, 2009; Wang et al., 2012).

Xiaochaihutang (XCHT), a famous Chinese herbal formula, was first recorded in “Shang Han Lun” two thousand years ago (Dai et al., 2008; Zhu et al., 2010). It consists of seven Chinese herbs: Chaihu (Radix Bupleuri), Huangqin (Radix Scutellariae), Renshen (Ginseng), Banxia (Rhizoma pinelliae), Gancao (Radix glycyrrhizae), Shengjiang (Rhizoma zingiberis recens) and Dazao (Fructus jujubae). XCHT has been used clinically in depressive disorders in China (Jia et al., 2009; Li and Gao, 1996). Recently, our previous study has demonstrated that XCHT significantly improved depressive-like behavior in several animal models of depression by alterations of serotonergic system and neurotrophic factors in hippocampus (Su et al., 2014a; Su et al., 2014b), which suggest that XCHT may be a potential candidate as a new therapy for depression.

However, the complexity of XCHT not only make identification of chemical constituents related to its antidepressant efficacy challenging, but also confuse the understanding of its action mechanisms on antidepressant therapy. Thus, a novel research strategy suitable for complex system is highly desired to provide comprehensive and further insight into the therapeutic basis and mechanism of XCHT.

In the present study, the core in compatibility of XCHT on antidepressant therapy was assessed by the method of orthogonal array design (Yi et al., 2009), and then series of behavioral tests, reserpine-induced pharmacological model were applied to estimate antidepressant effects of XCHT and its core in compatibility. The potential mechanisms were assessed by investigating levels of monoamine neurotransmitters and hippocampal neurogenesis. In addition, the identification and elucidation of active constituents in plasma by UPLC–MS/MS after oral administration of HRG were also studied in order to clarify the therapeutic basis of XCHT.

2. Materials and methods

2.1. Animals

Seven-week-old male Kunming mice weighing 18–22 g were supplied by the Experimental Animal Centre of Shenyang Pharmaceutical University. Animals were maintained on a standardized environmental conditions ($22 \pm 2^\circ\text{C}$, 12 h light/dark cycle) with free access to food and water and housed six per cage. All experiments were carried out according to the Regulations of Experimental Animal Administration issued by the State Committee of Science and Technology of China.

2.2. XCHT preparation

XCHT consists of seven Chinese herbs: 12 g of Radix Bupleuri (Chaihu; Latin name: *Bupleurum chinense* DC.), 9 g of Radix Scutellariae (Huangqin; Latin name: *Scutellaria baicalensis* Georgi), 9 g of Ginseng (Renshen; Latin name: *Panax ginseng* C.A. Meyer), 9 g of Rhizoma pinelliae (Banxia; Latin name: *Pinellia ternata* (Thunb.) Breit.), 6 g of Radix glycyrrhizae (Gancao; Latin name: *Glycyrrhiza uralensis* Fisch.), 6 g of Rhizoma zingiberis recens (Shengjiang; Latin name: *Zingiber officinale* Rosc.) and 9 g of Fructus jujubae (Dazao; Latin name: *Ziziphus jujuba* Mill.). All herbs were purchased from Tongrentang Chinese Pharmaceutical Co. Ltd. (Beijing, China) and identified by Professor Jiuzhi Yuan (School of Traditional Chinese Materia Medica, Shenyang Pharmaceutical University, China). Extracts of XCHT and other fourteen groups (Supplemental Table 1) were prepared by macerating the dried herbal

mixture in distilled water for 40 min, then boiling three times (100 g/1000 mL; 100 g/400 mL; 100 g/400 mL) for 30 min each time. The three decoctions were mixed and filtered, then lyophilized and stored in a desiccator. The yield of powdered extract was shown in Supplemental Table 1.

Reserpine (RES) was from Guangdong Bangmin Pharmaceutical Co. Ltd. (Guangdong, China). Amitriptyline (AMI) was from Hunan Dongting Pharmaceutical Co. Ltd. (Hunan, China).

2.3. Groups and drug administration

In part 1, to search the core in compatibility of XCHT on antidepressant therapy, mice were randomly divided into sixteen groups by the method of orthogonal array design ($n=12$). The selected doses for XCHT and other fourteen groups administration were based on our previous experiment (Su et al., 2014b). In part 2, to compare the antidepressant-like effects of XCHT and its core in compatibility, mice were randomly divided into eight groups ($n=22$). In part 3, to investigate the possible involvement of monoamine neurotransmitter mechanisms in the antidepressant-like effects of XCHT and its core in compatibility, reserpine-induced hypothermia and palpebral ptosis test was carried out, mice were randomly divided into nine groups ($n=12$). All procedures were shown in Fig. 1.

All drugs were dissolved in distilled water and given by gastric gavages once daily for 15 days. The vehicle mice were given with distilled water. Behavioral tests were conducted 1 h after 13, 14 and 15 days treatment, respectively.

2.4. Quality control of HRG and XCHT

3D-HPLC fingerprint of XCHT was performed as following procedure: Dried powder of XCHT (1.0 g) was extracted with methanol (10 mL) for 30 min under ultrasonication. Then the solution was centrifuged and the supernatant was filtered for HPLC analysis. HPLC was performed on a Shimadzu LC-10AD HPLC system equipped with a diode array detector and a LC workstation CLASS-VP (Shimadzu, Japan). Chromatographic conditions were shown in Supplemental material. The three-dimensional data results were shown in Fig. 2.

The contents of main components in HRG were further quantitatively analyzed. Lyophilized powder was dissolved in methanol with the concentration of 50 mg/mL. Subsequently, the solutions were centrifuged and the supernatant was filtered for the UPLC analysis. The UPLC–MS/MS system consisted of a Waters ACQUITY™ ultra performance liquid chromatography (UPLC) system (Waters Corp., Milford, USA) coupled to a Micromass Quattro Micro™ API mass spectrometer. Data were acquired and processed using MassLynx 4.1 software. The chromatographic separation and MS analysis were conducted according to the method described by previous study (Su et al., 2014a).

The contents of main components of HRG are Liquiritin (160 µg/g), Baicalin (6400 µg/g), Wogonoside (4060 µg/g), Baicalin (12 µg/g), Wogonin (1.7 µg/g), Oroxylin A (3.2 µg/g).

2.5. $L_{16}(2^7)$ orthogonal array design

XCHT was arranged by $L_{16}(2^7)$ orthogonal array design. Two levels of seven medicinal plants with a two-factor interaction were considered according to method described by (Yi et al., 2009). Briefly, seven factors were considered in this study: Chaihu (CH), Hangqin (HQ), Renshen (RS), Banxia (BX), Gancao (GC), Shengjiang (SJ), Dazao (DZ). In addition, twenty-one possible two-factor interactions (Supplemental Table 2) had also been examined.

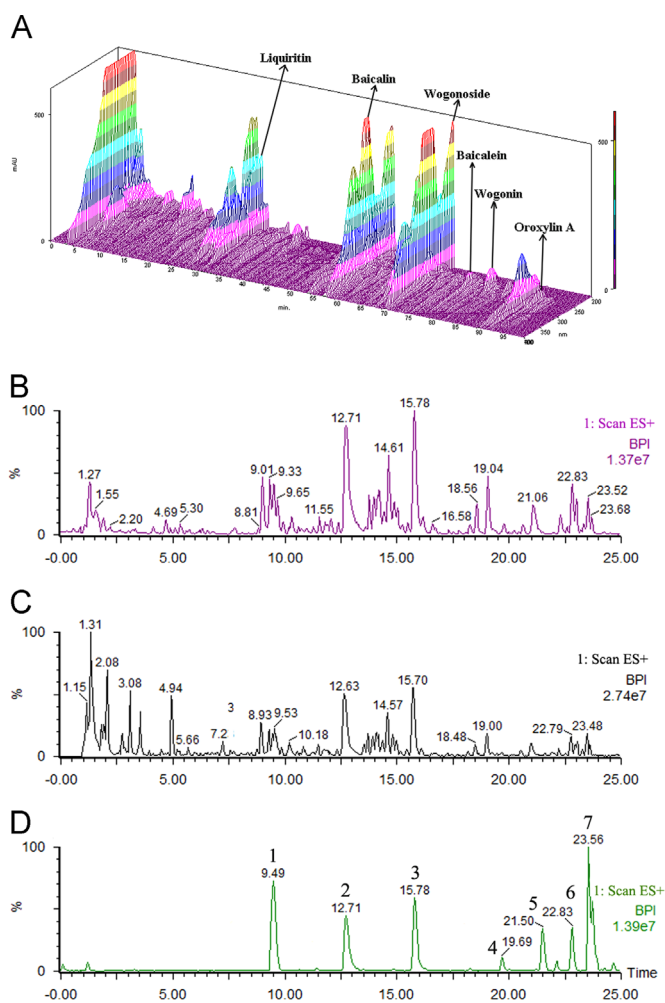


Fig. 2. 3D-HPLC fingerprint of XCHT (A) and UPLC total ion chromatograms of HRG (B); XCHT (C); and reference compounds (D). (1) Liquiritin, (2) Baicalin, (3) Wogonoside, (4) Baicalein, (5) Saikosaponin a, (6) Wogonin, (7) Oroxylin A.

2.6. Tail suspension test (TST) and forced swim test (FST)

Tail suspension test was performed according to the method described (Steru et al., 1985) with minor modifications. Briefly, mice were suspended on the edge of a metal rod 50 cm above a table top by adhesive tape placed approximately 1 cm from the tip of the tail. Testing was carried out in a darkened room to avoid background noise. Each mouse was suspended for a total of 6 min, and the immobility time was measured during the final 4 min of the test.

Forced swim test was used as previously described (Porsolt et al., 1978) with some modifications. Briefly, Mice were individually placed in plastic cylinder (height 40 cm, diameter 12 cm) containing 10 cm depth of water at $25 \pm 1^\circ\text{C}$ for 6 min. The immobility time was measured during the final 4 min of the test. The water was changed after each trial to avoid any influence of alarm substances. Both test sessions were recorded by a video camera and analyzed using Ethovision XT 8.0 software (Noldus, Wageningen, The Netherlands).

2.7. Locomotor activity

Locomotor activity was measured by an ambulator with eight activity chambers (Mobile Datum, Shanghai, CHN) using a modified method as previously reported (Li et al., 2004). The

system uses sensors with high-intensity, modulated infrared light beams to detect animal motion. Mice were placed into the cage activity system, the data-collecting system was immediately activated and locomotor activity was recorded for 10 min.

2.8. Novelty suppressed feeding test (NSFT)

NSFT has been formerly described (Bodnoff et al., 1989). Briefly, it consisted of a $30 \times 30 \times 25\text{ cm}^3$ box whose floor was covered with 2 cm sawdust. Animals were food deprived for 12 h before the test. At the time of testing, a single pellet of food was placed on a white paper positioned in the box center. Mice were placed in the corner and given 10 min to approach the food and eat. The latency to chew the pellet was recorded by an observer blind to this test. Immediately afterwards, animals were returned in their home cage, and the amount of food consumed by the mouse within 5 min was measured.

2.9. Reserpine-induced hypothermia and palpebral ptosis

According to the method described (Bourin et al., 1983), reserpine could induce hypothermia and palpebral ptosis. In this study, 30 min after final administration, mice intraperitoneally received reserpine (3 mg/kg). Rectal temperature and degree of palpebral ptosis were recorded at 1, 3, 5 and 7 h after reserpine treatment. The degree of palpebral ptosis was evaluated according to the following rating scale: (0) eyes open; (1) onequarter closed; (2) half closed; (3) three-quarters closed; (4) completely closed (Sanchez-Mateo et al., 2002). In addition, the rectal temperature of each animal was measured with a thermistor thermometer immediately after palpebral ptosis assessment.

2.10. Determination of neurotransmitter levels

Mice were sacrificed by the method of cervical dislocation and following decapitated, and brains were rapidly removed and dissected on an ice-chilled glass plate, including the hypothalamus and striatum. The tissue samples were weighed and stored at -80°C . Each frozen tissue sample was homogenized by ultrasonication in 200 μL of 0.4 M perchloric acid. The homogenate was kept on ice for 1 h and centrifuged at 13,000 rpm for 20 min. The supernatant was reserved and the contents of 5-HT, 5-HIAA, DA, DOPAC and HVA were measured by HPLC-ECD as described previously (Nitta et al., 1992). A LC-10A pump (Shimadzu, Kyoto, Japan) was set at 1.0 mL/min. The detector (L-ECD-6A, Shimadzu) was set at +0.70 V. The mobile phase consisted of 85 mM citrate, 100 mM sodium acetate, 0.2 mM EDTA, 1.2 mM sodium octanesulfonate and 15% methanol. The volume of injection was 20 μL .

2.11. Plasma sample preparation and qualitative analysis

Two hours after final administration of HRG (2.8 g/kg), blood samples were collected from the suborbital vein into heparinized tubes, and centrifuged at 4000 rpm for 5 min. The supernatant was obtained and stored at -80°C . All plasma samples from six mice per group were combined into one sample so as to eliminate the individual variability. A 400 μL plasma sample was extracted with protein precipitation with 800 μL acetonitrile. After centrifugation, 1 mL of supernatant was evaporated to dryness under a gentle stream of nitrogen. Then the residue was reconstituted with 100 μL of mobile phase (acetonitrile/water = 10/90), 10 μL of supernatant was injected into the UPLC-MS/MS system for analysis. Besides, the blank blood sample was processed at the same way. The plasma samples were detected according to the reported method relating to the chromatographic and MS conditions as described above.

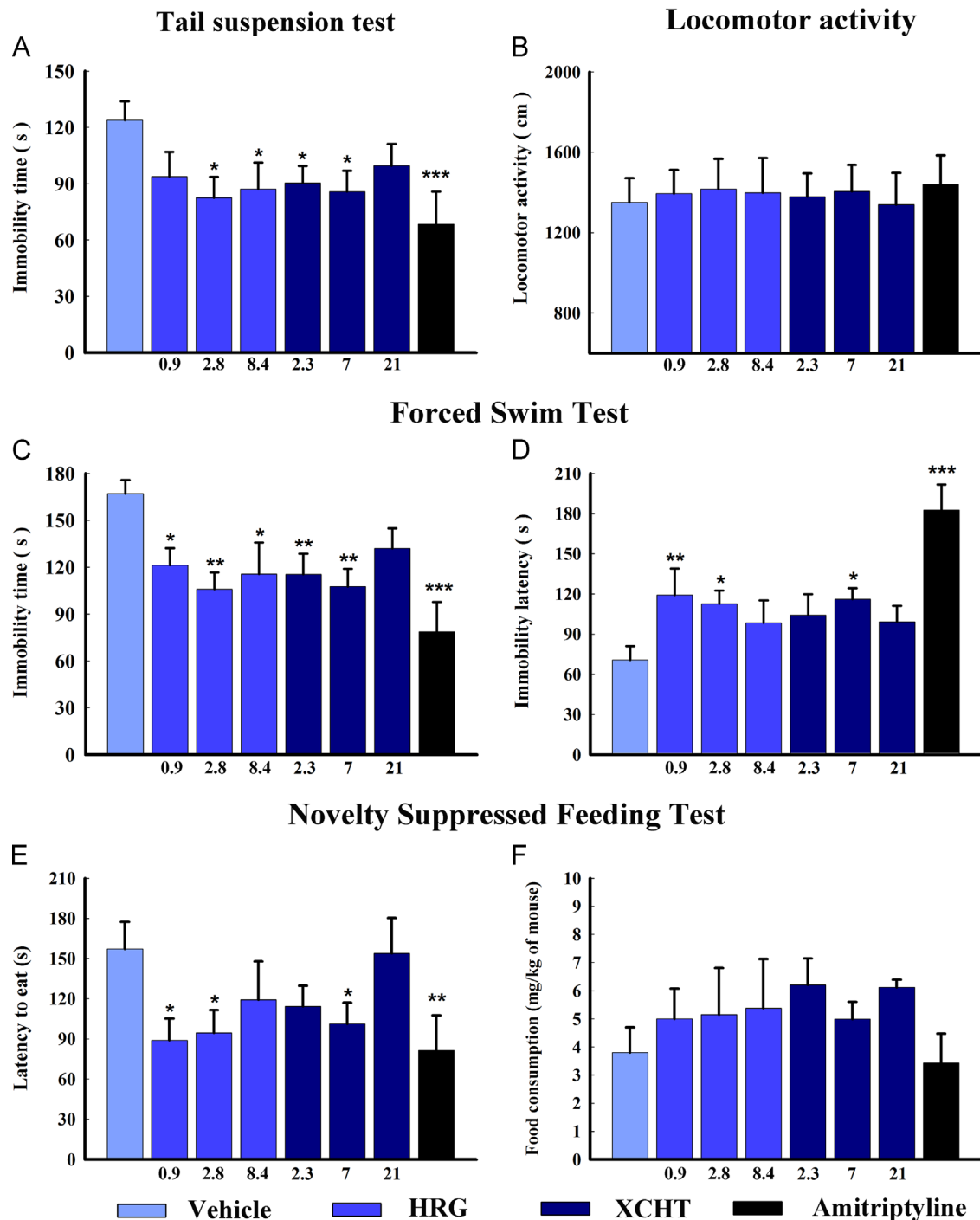


Fig. 3. Effects of HRG (0.9, 2.8 and 8.4 g/kg) and XCHT (2.3, 7 and 21 g/kg) on the TST (A), LA (B), FST (C and D) and NSFT (E and F). Data were expressed as means $\pm$ SEM ($n=12$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. vehicle.

2.12. Immunofluorescence

Mice were anesthetized by sodium pentobarbital (100 mg/kg, i. p.) and transcardially perfused through the heart with saline solution followed by 4% paraformaldehyde in phosphate buffer (0.1 M, pH 7.4). Brains were removed, post-fixed for 24 h in the same fixative, and transferred to 20% and 30% sucrose solution respectively at 4 °C. Then, the tissue was sectioned on a freezing microtome (AS-620, Shandon, Astmoor, UK) at a thickness of 20 μ m in three regions and stored at -20 °C allowing different immunofluorescence procedures. The immunofluorescence was conducted with Ki-67, DCX and NeuN as previous described

(Nollet et al., 2012). The sections were removed from -20 °C, adapted to room temperature, rehydrated and blocked with 10% normal goat serum for 1 h at 37 °C followed by incubating with rabbit monoclonal anti-Ki-67 antibody (1:1000, Abcam: ab15580), rabbit monoclonal anti-DCX antibody (1:500, Abcam: ab18723) and mouse polyclonal anti-NeuN antibody (1:1000, Millipore: mab377) diluted in 0.1 M PBS overnight at 4 °C. Secondary antibodies consisted of Cy3-conjugated goat anti-rabbit IgG (H+L) (1:200, Beyotime Institute of Biotechnology) and Alexa Fluor 488-conjugated goat anti-mouse IgG (H+L) (1:200, Beyotime Institute of Biotechnology) was used. Sections were incubated for 1 h at 37 °C and covered slips with anti-fade mounting medium

(Pro-Long Gold, Molecular Probes).

2.13. Statistical analysis

Data were expressed as the mean $\pm$ standard error of mean (SEM). In part 1, for orthogonal array design analysis of experiments, a two-way analysis of variance (ANOVA) was used. In part 2 and 3, comparisons between experimental and control groups were performed by one-way analysis of variance (ANOVA) followed by Dunnett's test. $P < 0.05$ was considered significant.

3. Results

3.1. The CORE in compatibility of herbs on antidepressant effect of XCHT

The result of orthogonal array design was shown in [Supplemental Table 1](#). After 15 days treatment, taking immobility time as index, ANOVA indicated that factor HQ, RS and GC were statistically significant, whereas the computed results of F indicated that any other factors and two-factor interactions were negligible and no statistical differences were observed ([Supplemental Tables 2 and 3](#)). So the formula consisted of Huangqin (*Radix scutellariae*), Renshen (*Ginseng*) and Gancao (*Radix glycyrrhizae*), which was defined as HRG, might be the core in compatibility of XCHT on antidepressant therapy.

3.2. Effects of HRG and XCHT on the immobility time in TST and FST

To investigate whether HRG was the core in compatibility of XCHT on antidepressant, the comparative evaluations on antidepressant effects of HRG and XCHT were executed by TST, FST. As shown in [Fig. 3A](#), oral administration of HRG (2.8 and 8.4 g/kg) and XCHT (2.3 and 7 g/kg) for 15 days significantly decreased the immobility time in TST compared to vehicle (all $P < 0.05$, the data of HRG and XCHT on day 7 were shown in supplementary material). [Fig. 3C](#) showed that HRG (0.9, 2.8, and 8.4 g/kg) and XCHT (2.3 and 7 g/kg) significantly decreased the immobility time in FST ($P < 0.05$ or $P < 0.01$). As seen in [Fig. 3D](#), HRG (0.9 and 2.8 g/kg) and XCHT (7 g/kg) induced a significant increase in immobility latency in FST ($P < 0.05$ or $P < 0.01$). AMI also significantly decreased the immobility time in TST and FST, with increased the immobility latency in FST ($P < 0.001$).

3.3. Effects of HRG and XCHT on locomotor activity and novelty suppressed feeding test

Effects of HRG and XCHT on Locomotor activity were shown in [Fig. 3B](#), HRG (0.9, 2.8, and 8.4 g/kg) and XCHT (2.3, 7, 21 g/kg) did not affect locomotor activity compared to vehicle (all $P > 0.05$). AMI did not affect locomotor activity ($P > 0.05$).

As shown in [Fig. 3E](#), the results showed that HRG (0.9 and 2.8 g/kg), XCHT (7 g/kg) and AMI significantly reduced the latency period until feeding in NSFT ($P < 0.05$ or $P < 0.01$), whereas without affecting food consumption were observed ([Fig. 3F](#)).

In accordance with XCHT, HRG exerts significant antidepressant effects in several animal models of depression. Next, the antidepressant effects of HRG by reserpine-induced hypothermia and palpebral ptosis test and the levels of monoamine neurotransmitters were evaluated.

3.4. Effects of HRG and XCHT on reserpine-induced hypothermia and palpebral ptosis

Mice treated with reserpine (3 mg/kg) showed significant alteration of hypothermia compared with vehicle at the 1, 3, 5 and 7 h ([Fig. 4A](#)). As shown in [Fig. 4B](#), pretreatment with HRG (0.9, 2.8, and 8.4 g/kg), XCHT (2.3, 7 and 21 g/kg) and AMI significantly reversed the hypothermia induced by reserpine at the 7 h compared to reserpine group (all $P < 0.001$). [Fig. 4C](#) showed that the degree of palpebral ptosis was significantly aggravated by reserpine treatment at the 7 h ($P < 0.001$). HRG (0.9, 2.8, and 8.4 g/kg), XCHT (2.3, 7 and 21 g/kg) and AMI significantly reversed palpebral ptosis induced by reserpine (all $P < 0.001$).

3.5. Effects of HRG and XCHT on monoamine neurotransmitter levels

According to results of behavioral tests, HRG at 2.8 g/kg and XCHT at 7 g/kg were the most effective dose, so we used that to analyze monoamine neurotransmitter levels. Effects of HRG, XCHT and AMI on levels of 5-HT, 5-HIAA, DA, DOPAC and HVA in hypothalamus and striatum were shown in [Table 1](#). In the hypothalamus, HRG (2.8 g/kg) and XCHT (7 g/kg) significantly increased 5-HT, DA and DOPAC levels ($P < 0.05$ or $P < 0.01$). Moreover, AMI also increased DA levels. In the striatum, HRG (2.8 g/kg) and XCHT (7 g/kg) significantly increased 5-HT and DOPAC levels ($P < 0.05$ or $P < 0.001$). Levels of 5-HIAA were decreased by HRG and XCHT ($P < 0.001$). Moreover, AMI significantly increased 5-HT, DA and DOPAC levels and decreased 5-HIAA, HVA.

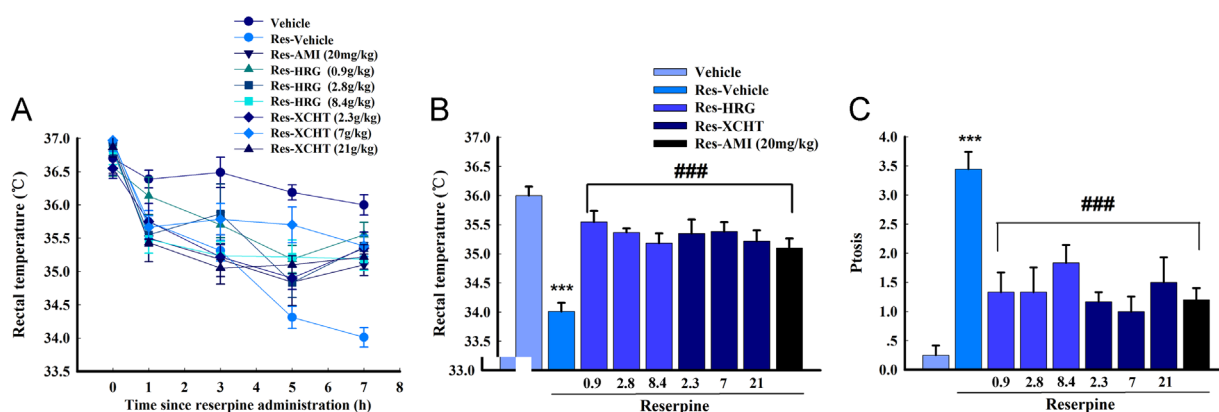


Fig. 4. A Representative image of HRG (0.9, 2.8 and 8.4 g/kg) and XCHT (2.3, 7 and 21 g/kg) on reserpine-induced hypothermia. [Fig. 4B](#) Effects of HRG (0.9, 2.8 and 8.4 g/kg) and XCHT (2.3, 7 and 21 g/kg) on reserpine-induced hypothermia at 7th hour. [Fig. 4C](#) Effects of HRG (0.9, 2.8 and 8.4 g/kg) and XCHT (2.3, 7 and 21 g/kg) on reserpine-induced palpebral ptosis at 7th hour. Data were expressed as means $\pm$ SEM ($n = 12$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. vehicle. ### $P < 0.001$ vs. reserpine.

Table 1
Effects of HRG and XCHT on monoamine neurotransmitter levels in hypothalamus and striatum.

Groups	Dose	5-HT	5-HIAA	DA	DOPAC	HVA
Hypothalamus						
Vehicle		336.2 ± 22.1	512.0 ± 19.9	207.8 ± 29.1	304.9 ± 11.4	274.6 ± 27.4
HRG	2.8 g/kg	515.2 ± 40.9***	552.6 ± 16.4	421.2 ± 16.4***	476.5 ± 23.1***	295.4 ± 32.7
XCHT	7 g/kg	494.1 ± 18.1**	533.4 ± 15.2	307.6 ± 20.6**	530.9 ± 21.9***	325.3 ± 17.3
AMI	20 mg/kg	414.8 ± 28.3	479.2 ± 20.8	295.8 ± 21.2*	310.7 ± 22.7	350.9 ± 27.7
Striatum						
Vehicle		336.8 ± 16.6	561.6 ± 12.3	3549.3 ± 248.6	4184.3 ± 207.6	2170.0 ± 160.1
HRG	2.8 g/kg	387.3 ± 14.1*	445.8 ± 11.4***	4136.2 ± 262.7	5331.7 ± 214.9***	1949.8 ± 100.4
XCHT	7 g/kg	403.1 ± 18.6*	487.3 ± 11.3***	4067.0 ± 219.8	4911.8 ± 290.6*	1938.6 ± 99.5
AMI	20 mg/kg	433.3 ± 12.9***	464.4 ± 12.8***	4295.4 ± 200.2*	5039.0 ± 235.6**	1458.1 ± 95.1***

Data were expressed as means ± SEM with units of ng/g. (n=8) vs. vehicle. 5-Hydroxytryptamine (5-HT); 5-Hydroxyindole-3-acetic acid (5-HIAA); Dopamine (DA); 3, 4-Dihydroxyphenylacetic acid (DOPAC); 4-Hydroxy-3-methoxyphenylacetic acid (HVA).

* $P < 0.05$

** $P < 0.01$

*** $P < 0.001$

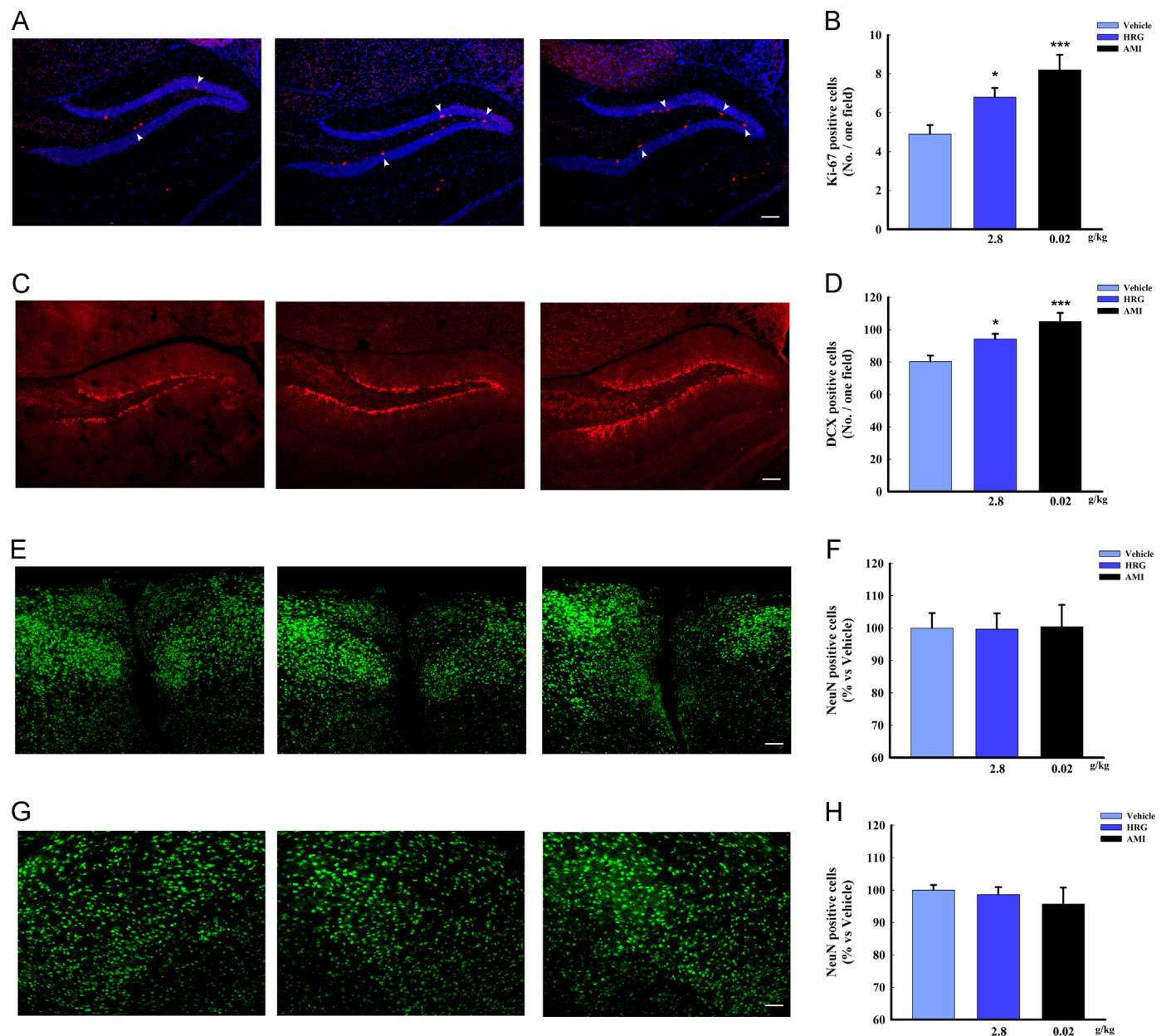


Fig. 5. Effects of HRG on neurogenesis (B and D) in dentate gyrus of hippocampus. Representative images of Ki-67-positive (white arrowheads) and DCX-positive in hippocampus were shown (A and C). NeuN immunofluorescence stained in hypothalamus (F) and striatum (H). Representative images of NeuN-positive in hypothalamus and striatum were shown (E and G). Scale bar: 100 μ m. Data were expressed as means ± SEM (n=10/group). * $P < 0.05$, *** $P < 0.001$ vs. vehicle.

3.6. Effects of HRG on neurogenesis in hippocampus and neuronal loss in hypothalamus and striatum

Effects of HRG on neurogenesis in hippocampus were shown in Fig. 5A–D. Oral administration of HRG (2.8 g/kg) for 15 days could significantly increase Ki-67 and DCX positive cells ($P < 0.05$). In addition, NeuN was used to assess neuronal loss in hypothalamus and striatum (Fig. 5E–H), and no significant difference in the number of neuron was found ($P > 0.05$).

3.7. UPLC–MS/MS analysis of plasma sample after oral administration of HRG

Plasma samples after oral administration of HRG were analyzed by UPLC–MS/MS method. A total 25 compounds were identified, including 14 prototype components and 11 metabolites. Extracted

ion chromatograms of blank plasma sample and plasma samples after administered HRG were shown in Fig. 6. The retention time (t_R) and MS data of the metabolites were listed in Table 2.

There were 14 peaks (marked as P1–P14) displayed in the profiles of dosed plasma and HRG whereas there were no equivalent peaks in the profile of the blank plasma. Thus, these compounds were defined as prototype components. According to the retention times and mass spectra with those of authentic compounds, peaks 1, 2, 3, 4, 9, 11, 12, 13 and 14 were designated as Liquiritin, Ginsenoside Rg1, Ginsenoside Re, Baicalin, Wogonoside, Baicalein, Glycyrrhizic acid, Wogonin, Oroxylin A, respectively. Peaks 5, 6, 7, 8 and 10 were tentatively characterized as OroxylinA-7-O-glu acid, Trihydroxy-methoxy-glu acid flavones, Chrysin-glu acid and Baicalein-6-O-glu acid, respectively. The other 11 peaks marked as M1–M11, which only appeared in dosed plasma, were assumed to be exogenous metabolites derived from HRG.

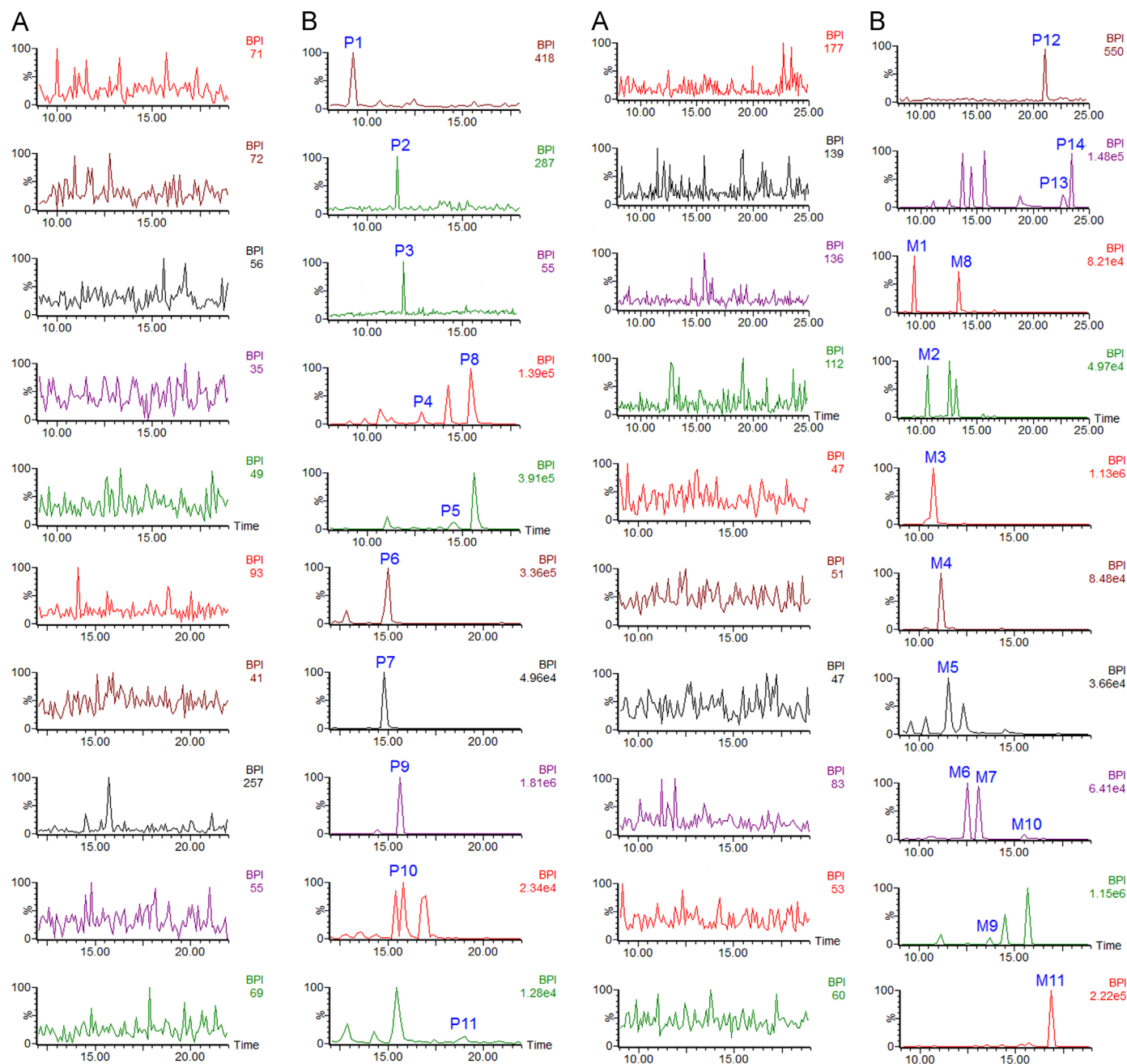


Fig. 6. Extracted ion chromatograms of (A) blank mice plasma and (B) mice plasma sample 2 h after oral administration of HRG.

Table 2

The prototype components and metabolites detected from plasma of mice after administration of HRG.

NO.	t _R /min	MS (m/z)	MS/MS (m/z)	Identification
P1	9.27	441 [M+Na] ⁺	257 [M+H-Glu] ⁺	Liquiritin ^a
P2	11.62	823 [M+Na] ⁺	239 [M+H-Glu-H ₂ O] ⁺ 621 [M+H-Glu-H ₂ O] ⁺ 603 [M+H-Glu-2H ₂ O] ⁺ 423 [M+H-2Glu-3H ₂ O] ⁺ 405 [M+H-2Glu-4H ₂ O] ⁺	Ginsenoside Rg1 ^a
P3	11.82	970 [M+Na] ⁺	749 [M+H-Glu-2H ₂ O] ⁺ 587 [M+H-2Glu-2H ₂ O] ⁺ 441 [M+H-2Glu-2H ₂ O-Rha] ⁺	Ginsenoside Re ^a
P4	12.85	447 [M+H] ⁺ 469 [M+Na] ⁺	271 [M+H-GluA] ⁺	Baicalin ^a
P5	14.42	461 [M+H] ⁺ 483 [M+Na] ⁺	285 [M+H-GluA] ⁺ 270 [M+H-GluA-CH ₃] ⁺	OroxylinA-7-O-glu acid
P6	14.80	477 [M+H] ⁺ 499 [M+Na] ⁺	301 [M+H-GluA] ⁺	Trihydroxy-methoxy-glu acid flavone
P7	14.81	431 [M+H] ⁺	255 [M+H-GluA] ⁺	Chrysin-glu acid
P8	15.43	447 [M+H] ⁺ 469 [M+Na] ⁺	271 [M+H-GluA] ⁺	Baicalein-6-O-glu acid
P9	15.64	461 [M+H] ⁺ 483 [M+Na] ⁺	285 [M+H-GluA] ⁺	Wogonoside ^a
P10	15.79	491 [M+H] ⁺ 513 [M+Na] ⁺	315 [M+H-GluA] ⁺	Dihydroxy-dimethoxy-glu acid flavone
P11	19.02	271 [M+H] ⁺	253 [M+H-H ₂ O] ⁺ 169 [M+H-C ₈ H ₆] ⁺	Baicalein ^a
P12	21.03	823 [M+H] ⁺ 845 [M+Na] ⁺	647 [M+H-GluA] ⁺ 471 [M+H-2GluA] ⁺ 453 [M+H-2GluA-H ₂ O] ⁺	Glycyrrhizic acid ^a
P13	22.61	285 [M+H] ⁺	270 [M+H-CH ₃] ⁺ 168 [M+H-CH ₃ -C ₈ H ₆] ⁺	Wogonin ^a
P14	23.41	285 [M+H] ⁺	270 [M+H-CH ₃] ⁺ 168 [M+H-CH ₃ -C ₈ H ₆] ⁺	Oroxylin A ^a
M1	9.39	433 [M+H] ⁺ 455 [M+Na] ⁺	257 [M+H-GluA] ⁺ 239 [M+H-GluA-H ₂ O] ⁺ 163 [M+H-GluA-C ₆ H ₆ O] ⁺	Liquiritigenin-glu acid
M2	10.58	507 [M+H] ⁺ 529 [M+Na] ⁺	331 [M+H-GluA] ⁺ 316 [M+H-GluA-CH ₃] ⁺ 301 [M+H-GluA-2CH ₃] ⁺	Trihydroxy-dimethoxy-glu acid flavone
M3	10.77	623 [M+H] ⁺ 645 [M+Na] ⁺	447 [M+H-GluA] ⁺ 271 [M+H-2GluA] ⁺ 169 [M+H-2GluA-C ₈ H ₆] ⁺	Baicalein-diglu acid
M4	11.15	637 [M+H] ⁺ 659 [M+Na] ⁺	461 [M+H-GluA] ⁺ 285 [M+H-2GluA] ⁺ 270 [M+H-2GluA-CH ₃] ⁺	Wogonin- diglu acid
M5	11.54	653 [M+H] ⁺	477 [M+H-GluA] ⁺ 301 [M+H-2GluA] ⁺	Trihydroxy-methoxy-diglu acid flavone
M6	12.52	675 [M+Na] ⁺ 507 [M+H] ⁺ 529 [M+Na] ⁺	286 [M+H-2GluA-CH ₃] ⁺ 331 [M+H-GluA] ⁺ 316 [M+H-GluA-CH ₃] ⁺ 301 [M+H-GluA-2CH ₃] ⁺	Trihydroxy-dimethoxy-glu acid flavone
M7	13.12	507 [M+H] ⁺ 529 [M+Na] ⁺	331 [M+H-GluA] ⁺ 316 [M+H-GluA-CH ₃] ⁺ 301 [M+H-GluA-2CH ₃] ⁺	Trihydroxy-dimethoxy-glu acid flavone
M8	13.36	433 [M+H] ⁺ 455 [M+Na] ⁺	257 [M+H-GluA] ⁺ 239 [M+H-GluA-H ₂ O] ⁺	Isoliquiritigenin-glu acid
M9	13.71	461 [M+H] ⁺ 483 [M+Na] ⁺	285 [M+H-GluA] ⁺ 270 [M+H-GluA-CH ₃] ⁺	Wogonin-5-O-glu acid
M10	15.50	507 [M+H] ⁺ 529 [M+Na] ⁺	331 [M+H-GluA] ⁺ 316 [M+H-GluA-CH ₃] ⁺ 301 [M+H-GluA-2CH ₃] ⁺	Trihydroxy-dimethoxy-glu acid flavone
M11	16.88	491 [M+H] ⁺ 513 [M+Na] ⁺	315 [M+H-GluA] ⁺ 300 [M+H-GluA-CH ₃] ⁺ 285 [M+H-GluA-2CH ₃] ⁺	Dihydroxy-dimethoxy-glu acid flavone

^a Compounds identified by reference standards.

M1 and M8 showed the same [M+H]⁺ ion at *m/z* 433 and afforded fragment ions at *m/z* 239 and 169, respectively. By referring to the literature data, they were tentatively identified as Liquiritigenin-glu acid and Isoliquiritigenin-glu acid (Qiao et al., 2012). In addition, M2, M6, M7 and M10 exhibited the same [M+H]⁺ ions at *m/z* 507 and [M+Na]⁺ ions at *m/z* 528, and they also displayed the identical fragment ions at *m/z* 331, 316 and 301,

respectively. Whereas, the retention times of them were different. They were tentatively identified as the isomers of Trihydroxy-dimethoxy-glu acid flavone. The identification and structural elucidation of other metabolites were shown in Table 2.

4. Discussion

In China, XCHT is an effective Chinese herbal formula, which

has been used for treating depressive disorders in clinical situations (Jia et al., 2009; Li and Gao, 1996). In the present study, the core in compatibility of XCHT on antidepressant therapy was evaluated by orthogonal array design, which provides an efficient testing strategy in biological study (Guo et al., 2007) and is generally used to study pharmacological compatibility of prescriptions (Yi et al., 2009). After 15 days treatment, the result of compatibility study showed that HQ, RS and GC significantly reduced the immobility time in both TST and FST. Therefore, HRG might be the core in compatibility of XCHT on antidepressant therapy. To verify this result, antidepressant effects of HRG and XCHT were further compared by TST, FST, NSFT and reserpine-induced hypothermia and ptosis. It is well known that TST and FST represent two behavioral despair models, which are the most widely used for evaluating antidepressant activity (Steru et al., 1985). In the present study, in accordance with XCHT, HRG (2.8 and 8.4 g/kg) significantly reduced the immobility time in both TST and FST. In addition, drugs influencing locomotor activity may give a false positive effect in TST and FST (Borsini and Meli, 1988). In order to rule out this possibility, locomotor activity was measured and no statistical differences were observed. These results suggested that HRG decreased the immobility time in behavioral despair tests was not related to a psychostimulant effect, but rather an antidepressant effect. The NSFT is a conflict test that elicits competing motivations: the drive to eat and the fear of venturing into the center of a bright arena. The latency to eat, an index to evaluate depression related behaviors, is decreased by chronic antidepressant drug treatments. In our study, both HRG (0.9 and 2.8 g/kg) and XCHT (7 g/kg) significantly reduced the latency period to eat in NSFT without affecting appetite of mice. As we know, depression has been associated with disturbances of brain monoamine neurotransmitters. Reserpine was shown to irreversibly inhibit the vesicular uptake of monoamines, including NE, 5-HT, DA and its metabolites, while HRG and XCHT significantly reversed reserpine-induced hypothermia and ptosis. The result indicated that brain monoamine neurotransmitters might be involved in HRG and XCHT mediated alleviation of depressive-like behavior. Overall, HRG was quite equivalent to XCHT in series of behavioral tests and might be the core in compatibility of XCHT on antidepressant therapy.

Most clinical researches demonstrated serotonergic system and dopaminergic system played major roles in depression (Hirschfeld, 1999; Robinson et al., 2012). Our previous study has demonstrated that XCHT enhances the serotonergic system in the prefrontal cortex and hippocampus (Su et al., 2014b). In order to further verify the antidepressant-like effects of HRG and XCHT, 5-HT, DA and their metabolites in hypothalamus and striatum were detected by HPLC-ECD after 15 days treatment. Our result showed that both HRG (2.8 g/kg) and XCHT (7 g/kg) significantly increased 5-HT levels in the hypothalamus and striatum. In addition, both HRG and XCHT significantly increased DA and DOPAC levels in hypothalamus and striatum. This result was in accordance with our previous study that XCHT significantly increased DA level in the hippocampus in a rat model of chronic unpredictable mild stress (Su et al., 2014a). Overall, all the above data suggested that HRG might exert antidepressant effects via enhancing the activity of serotonergic system and reward pathway of dopaminergic system.

Hippocampus, one of the brain regions, plays the key role in regulating the emotion and cognition (Moser and Moser, 1998). A particular attention was focused on hippocampus, based on the facts that most antidepressants increased hippocampal neurogenesis, and blocking hippocampal neurogenesis could prevent part of antidepressant effects in rodents (David et al., 2009; Santarelli et al., 2003). Our previous study demonstrated that XCHT significantly increased BDNF, NGF and their receptors (TrkB and

TrkA) expression in the hippocampus (Su et al., 2014a), and these neurotrophic factors might contribute to hippocampal neurogenesis. Based on above reasons, it was speculated that hippocampal neurogenesis might be involved in antidepressant-like effect of HRG. In accordance with XCHT, our result showed that HRG could significantly increase hippocampal neurogenesis. Moreover, neuroinflammation is one of hypotheses that inflammatory processes and brain-immune interactions are involved in the pathogenesis of major depression, and may contribute to the dysfunction of monoamine neurotransmitters (Song and Wang, 2011). The excessive inflammatory responses have been reported to contribute to neuronal apoptosis and death (Woodman and Lockette, 2009), so it is essential to show neuronal loss may be found in hypothalamus and striatum. In our study, NeuN was used to assess neuronal loss in hypothalamus and striatum, but no significant difference of HRG in the number of neuron in hypothalamus and striatum was found. Moreover, in our further studies, the role of HRG in neuroinflammation and depression will be comprehensively and deeply confirmed.

Traditional Chinese medicine is an essential part of herbal medicine. It has been considered that the therapeutic effects of herbal medicine are due to the interaction of individual chemical components and multiple targets. To clarify the active constituents responsible for the pharmacological action, we used an UPLC–MS/MS method to reveal the chemical constituents profile in vivo. In total, 25 constituents from plasma were identified or tentatively characterized after oral administration of HRG. Among them, 14 constituents were untransformed constituents and the other 11 constituents were biotransformed metabolites. According to previous report, Baicalin, as the representative component of *Radix Scutellariae*, significantly reduced immobility time in TST and FST in mice by mediating in part through MAO A and B inhibition (Zhu et al., 2006). In addition, Ginsenoside Rg1 (P2) had a protective effect on chronic restraint stress, activated BDNF signaling pathway and increased neurogenesis in the hippocampus (Wang et al., 2014). Liquiritin (P1) exerted antidepressant effect in chronic variable stress induced depression model by the mechanism of antagonizing oxidative stress (Zhao et al., 2008). All constituents derived from *Radix Scutellariae*, Ginseng and *Radix Glycyrrhizae* also indicated that HRG might be the core in compatibility from XCHT. These constituents might be the potential active ingredients to treat depression. However, more targets of antidepressant effects of HRG and XCHT need to be compared, and then chemical profile of active constituents in brain after oral administration of HRG will be investigated in future studies.

5. Conclusion

This is the first report that the formula, which consist of *Radix scutellariae*, Ginseng and *Radix glycyrrhizae*, named as HRG, is the core in compatibility of XCHT on antidepressant therapy through tail suspension test, forced swim test, novelty suppressed feeding test, reserpine test. Furthermore, the present results indicate that HRG is quite equivalent to XCHT in antidepressant effect, which are likely attributed to regulating the serotonergic and dopaminergic systems and increasing hippocampal neurogenesis. In addition, the identification and elucidation of active constituents in plasma after oral administration of HRG provide essential data for the pharmacodynamic profile, which contribute to clarifying the therapeutic basis of XCHT and facilitating its clinical usage.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.jep.2015.08.031>.

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Corrigendum

Corrigendum to “Analysis of main constituents and mechanisms underlying antidepressant-like effects of Xiaochaihutang in mice” [J. Ethnopharmacol. 175 (2015) 48–57]



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The authors regret that “L₁₆ (2⁷)” should be replaced by “L₁₆ (2¹⁵)” in Materials and methods and Supplementary material. The authors would like to apologise for any inconvenience caused.

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